

EVALUATING THE POTENTIAL IMPACT OF WA DNR RECREATION CLOSURES

Status Report #2

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TIMELINE

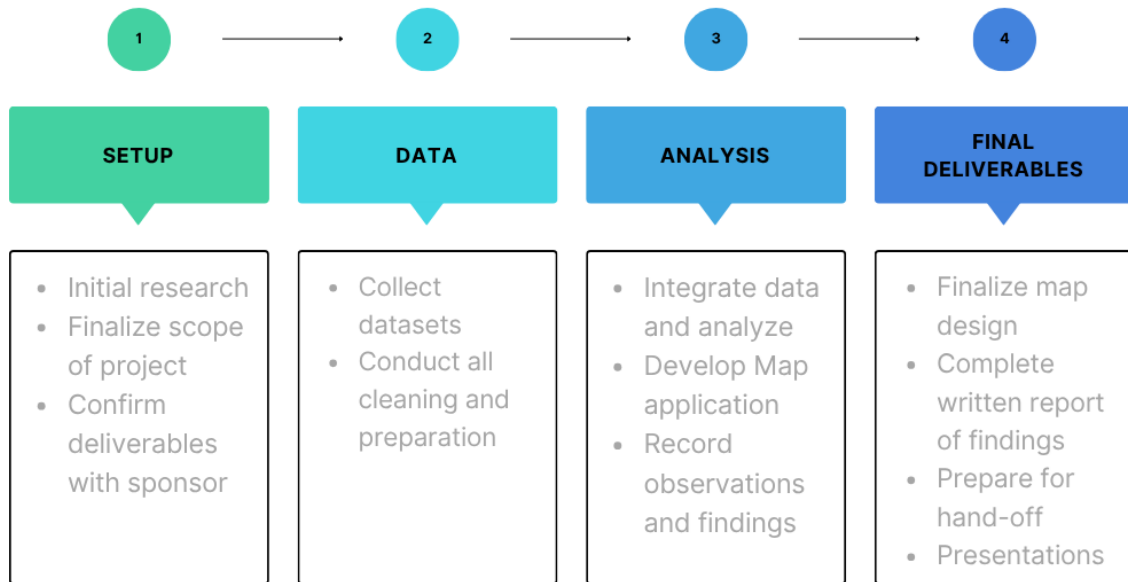
Milestone 1: Complete

Milestone 2: Complete

Milestone 3: Finalizing

Milestone 4: Upcoming

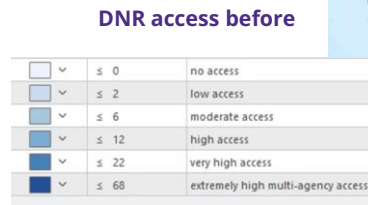
Project Milestones



PROGRESS TO DATE

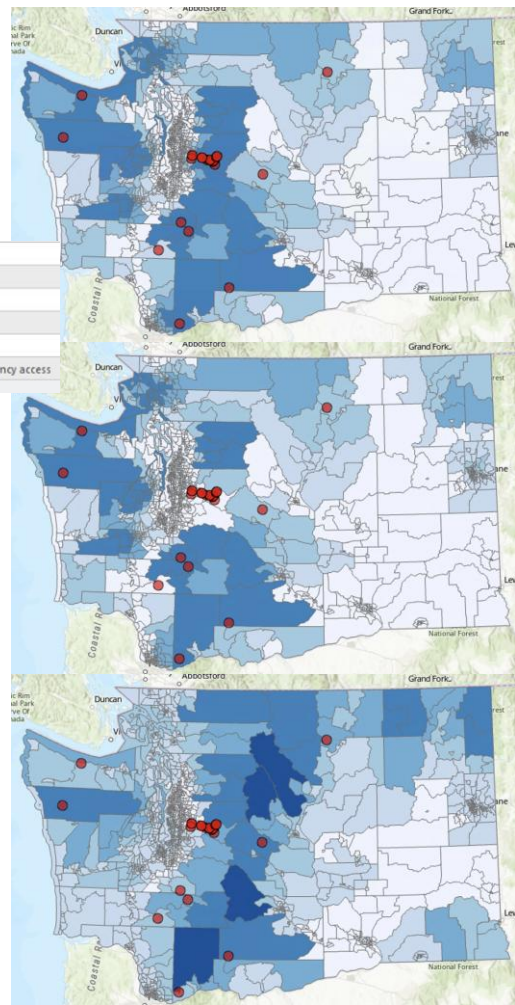
Recreation Accessibility & Demographic Impact Analysis

- Modeled recreation accessibility using 10-mile proximity buffers around recreation sites
- Used tract-level Join_Count analysis to measure nearby recreation opportunities
- Accessibility patterns strongly shaped by geography
 - Rural tracts generally remained closer to public lands
 - Urban and suburban tracts outside modeled buffers
- Added other agencies beyond DNR (USFS and Washington State Parks sites)
 - Multiagency increased number of accessible tracts from about 550 to over 1,100 tracts
 - 651 tracts (3M people) still have zero access
- Considerations
 - Accessibility modeled as nearby recreation opportunities within 10 miles
 - Accessibility also shaped by transportation, recreation behavior, and financial barriers



DNR access after

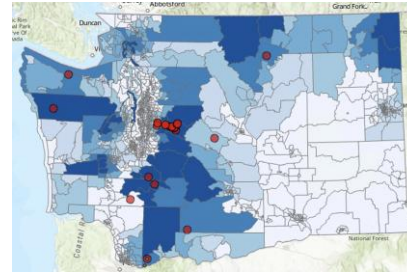
After with multi-agencies



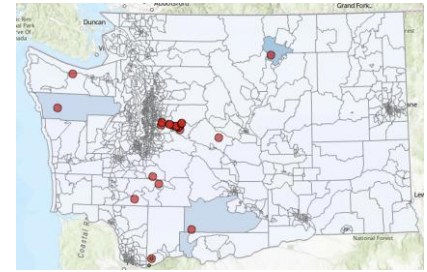
Activity-Specific Accessibility

- Examined activity-specific accessibility vs treating all recreation sites as interchangeable
- Accessibility impacts vary depending on recreation type and availability of nearby substitution
 - Hiking and scenic viewing experienced some of the largest accessibility reductions
 - Hiking retained some substitute networks after closures
 - Scenic viewing became more fragmented with fewer overlapping opportunities

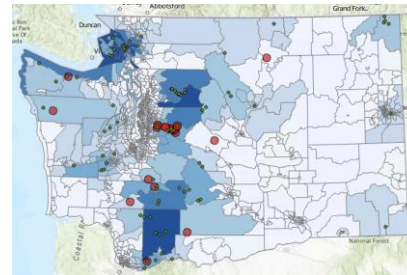
Hiking access before



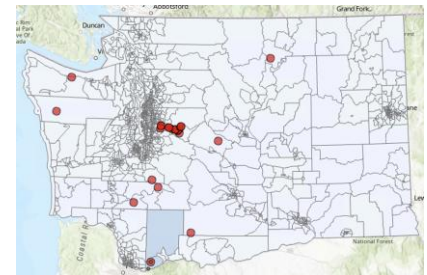
Hiking access after



Scenic viewing access before



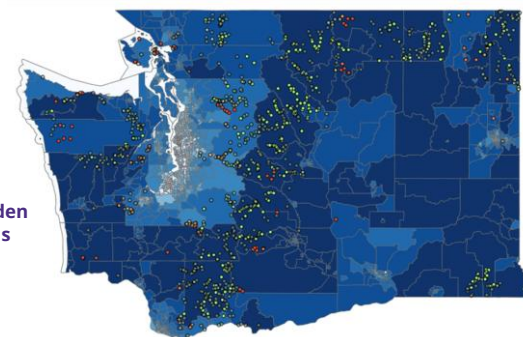
Scenic viewing access after



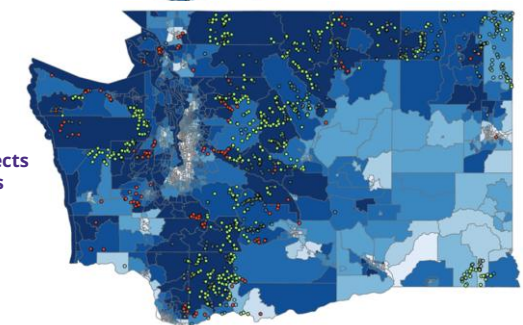
Environmental / Equity Overview

- Joined DNR & USFS point data with WA census tract Health disparity data
 - Environmental Exposure & Effects, Socioeconomic Factors, Sensitive Populations
 - Site info: location, amenities, activities
- Assign each location with local environmental and socioeconomic conditions
 - Describe local population being served
 - Assess options provided for peoples undergoing different challenges and health burdens
 - Compare conditions at locations with similar services
- Provide alternative perspective on value per recreation site
- Note: Disparity information accurate at census tract level

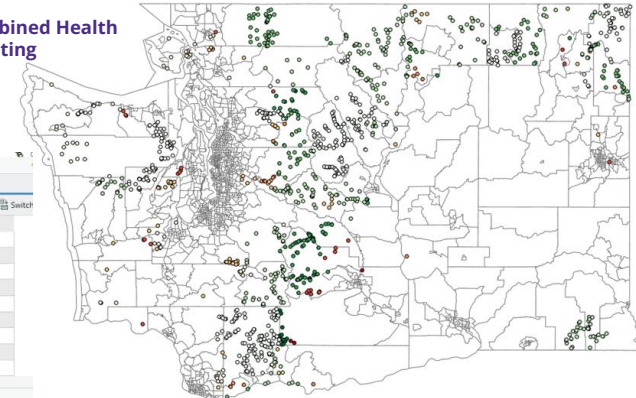
Map: Transport Burden
Points: Campgrounds



Map: Environmental Effects
Points: Hiking/trailheads



Points: Combined Health Disparity Rating

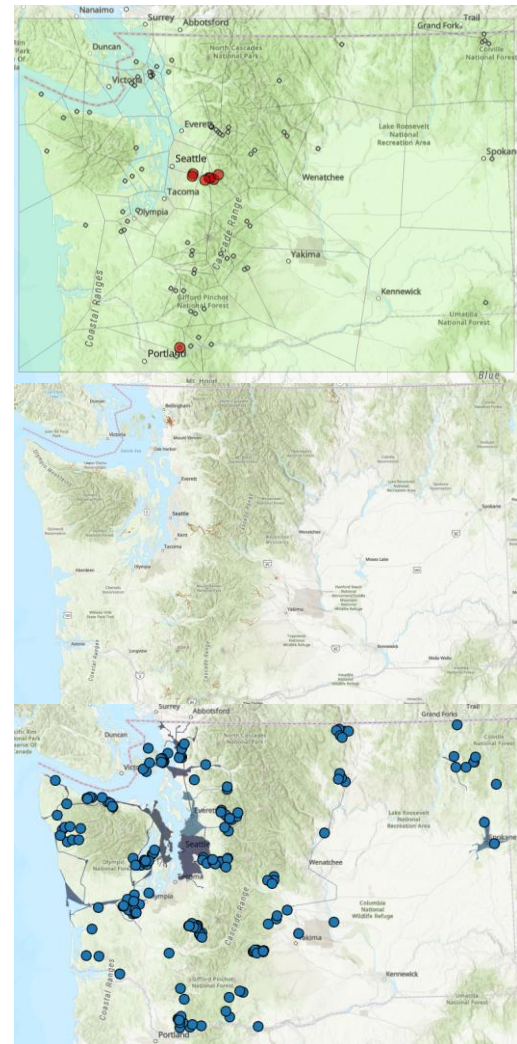


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Panoramic	0	0	0	0	0	0	0	0	0
Rock Climbing	0	0	0	0	0	0	0	0	0
Scenic Viewing	0	0	0	0	0	0	0	0	0
Wildlife Viewing	0	0	0	0	0	0	0	0	0
TRAC10	0	0	0	0	0	0	0	0	0
GEOD10	0	0	0	0	0	0	0	0	0
CountyFIPS	0	0	0	0	0	0	0	0	0
County10	0	0	0	0	0	0	0	0	0
Dispersal Rank	0	0	0	0	0	0	0	0	0
County Rank	0	0	0	0	0	0	0	0	0
PM2.5 Rank	0	0	0	0	0	0	0	0	0
Proximity1	0	0	0	0	0	0	0	0	0
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OBJECTID	1	1	84	0.035766	1	DNR
IN_FID	2	1	664	0.036216	2	USFS
NEAR_FID	3	1	23	0.0387	3	DNR
NEAR_DIST	4	1	209	0.048878	4	DNR
NEAR_RANK	5	1	8	0.046635	5	DNR
NEAR_FC	6	6	2	0.05263	1	USFS
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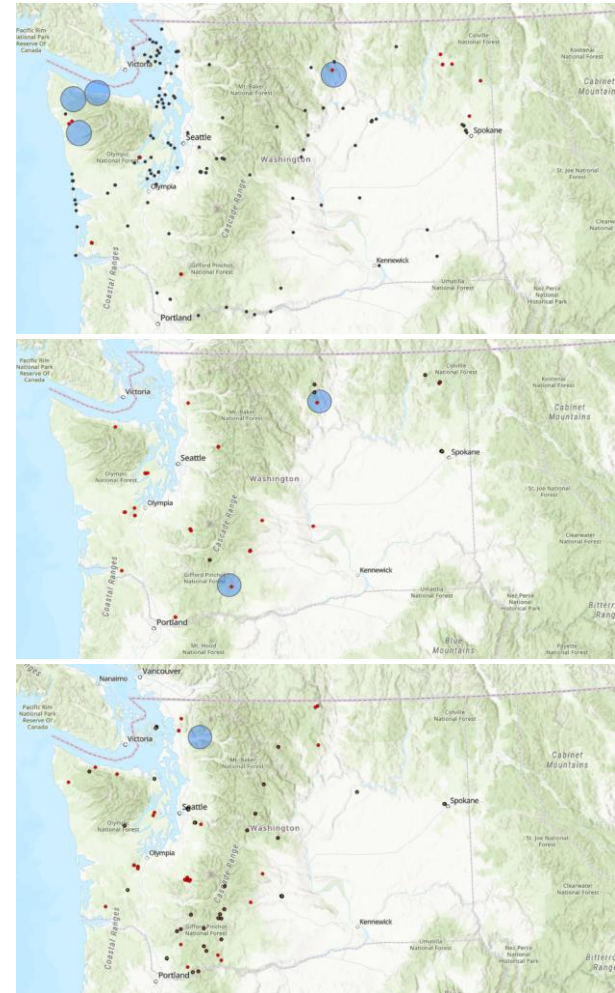
Site Area/Alternative Analysis

- Objective: analyze which areas were the most impacted by site closures in terms of loss of activities
- Initial goal: analyze changes in service area before/after site closures to measure changes in access
- Methods: drive time analysis, public transit time analysis, thiesen polygons
- Challenges: public transit analysis too niche; drive time faulty; thiesen polygons too confusing and convoluted
- Changing objectives: after confirming impacted and closed sites, identify alternative recreational activity sites



Further Alternative Site Analysis

- Fully Closed sites
 - Alternative Options for Fishing OHVs/ATVs and Horseback Riding
 - Activities with fully closed sites excluding Snoqualmie closures
- USFS/Parks/DNR Alternatives
 - 10 mile buffers
 - Not many options with especially for the Olympic peninsula fishing, more often DNR alternatives are best
 - ATVs in more concentrated areas and will have DNR alternatives
 - Plenty of Fishing and Horseback Riding across the state
 - Effort must be made to leave the 10 mile zone for Equestrian activities



Identifying Alternative Sites

- New goal: identify closest alternative sites that provide the activity lost for each closed/impacted site
- Method: split all sites by activity and create analyses for each activity
- Create a layer of all open sites (WA DNR, USFS, WA State Parks) and a layer of closed DNR sites with the same activities
- Use Near tool to identify alternative open sites with the same activity within 10 miles

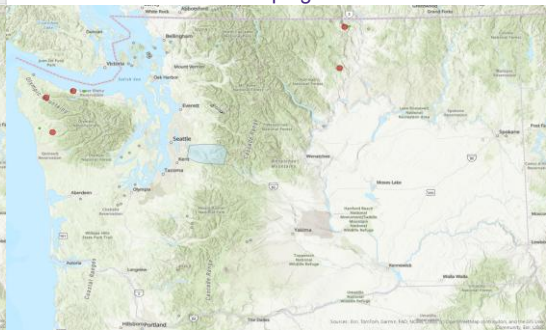
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Camping



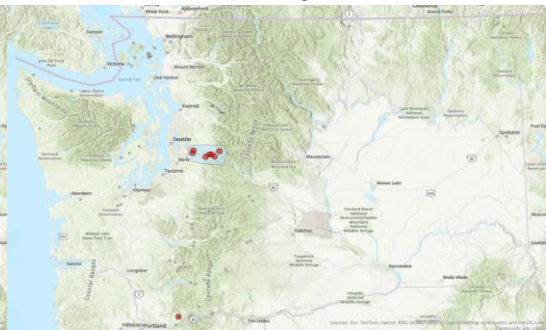
Hiking



Fishing



Kayaking



Scenic Viewing